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Real-time Intelligent Identification of UAV RF Fingerprints Based on YOLO26 and Embedded System Validation

Hanqing Xue†, Rongkai Qie†, Chao Chen†, Qibin He, Peng Wang, Xiaoming Dou, Lingxi Guo and Haiwen Sun*

Abstract— **With the proliferation of unmanned aerial vehicles (UAVs), non-cooperative target detection has become a critical security concern. Traditional radar and optical methods often fail to detect small-scale UAVs in complex environments. This paper proposes a novel RF fingerprint identification technique leveraging the YOLO26 architecture. By transforming raw I/Q signals into high-resolution time-frequency spectrograms, we treat signal identification as a specialized object detection task. The YOLO26 model is optimized with the loss function to improve sensitivity to narrow-band frequency hopping signals. Experimental results demonstrate that the proposed method achieves a Mean Average Precision (mAP50) of 98.1% on public datasets. Furthermore, the system delivers a real-time inference speed of 21.73 FPS on the Huawei Ascend 310B AI chip, which verifies its effectiveness for edge deployment in low-altitude security scenarios.**

I. Introduction

The rapid proliferation and technological advancement of Unmanned Aerial Vehicles (UAVs) have significantly contributed to various industrial applications; however, the escalating prevalence of unauthorized UAVs operations poses substantial threats to low-altitude airspace security and public safety. Efficient and accurate detection of non-cooperative UAVs is essential for maintaining UAVs Traffic Management (UTM) security [1].

Traditional surveillance methods exhibit inherent limitations in complex urban and low-altitude environments. Radar systems often lack the sensitivity required to detect small-scale UAVs, which are predominantly constructed from plastic or carbon fiber materials. Similarly, acoustic detection techniques are highly susceptible to environmental noise interference, leading to a high rate of false alarms and limited detection range.

In contrast, RFF offers a high-reliability, non-line-of-sight (NLOS) detection alternative. The radio control and telemetry signals emitted by UAVs serve as a unique "identification card" [2]. By leveraging deep learning architectures to analyze the time-frequency spectrograms of these signals, it is possible to achieve robust identification of specific UAV models even in cluttered electromagnetic environments [3].

†Equal contribution. ∗Corresponding author.
Hanqing Xue, Rongkai Qie, Chao Chen, Qibin He, Peng Wang, Xiaoming Dou and Lingxi Guo are with the National Key Laboratory of Nearspace Physics, Beijing 100076, China(e-mail: 982132969@qq.com).
Corresponding author Haiwen Sun is with Research Institute, Beijing 100076, China(e-mail: 842904820@qq.com).

The primary significance of this research lies in the application of the state-of-the-art YOLO26 algorithm [4] to solve the critical trade-off between recognition accuracy and real-time performance. Existing deep learning models often encounter deployment difficulties on resource-constrained embedded systems due to their computational intensity. This study introduces an optimized YOLO26-based framework designed for intelligent multi-model identification. Our approach aims to provide essential technical support for low-altitude security defense systems by ensuring high-fidelity detection and seamless edge-side deployment in practical, complex electromagnetic scenarios.

II. Research Status

The widespread deployment of Unmanned Aerial Vehicles (UAVs) in civilian and military sectors has introduced significant security risks, including identity spoofing, illegal reconnaissance, and privacy infringement [1, 5]. Traditional vision and video processing techniques often exhibit limited performance under low-light conditions, high-dynamic positioning fluctuations, and the absence of global positioning information [5]. To address these constraints, Radio Frequency Fingerprinting (RFF) has emerged as a robust physical-layer identification technology. RFF exploits minute, nearly unforgeable hardware impairments—such as oscillator instability, power amplifier nonlinearity, and I/Q imbalance—generated during the manufacturing process. Compared to conventional bit-level security like IP/MAC address encryption, RFF provides a hardware-rooted authentication mechanism that is exceptionally resistant to malicious imitation [6].

Early research in UAVs detection primarily utilized neural networks to automate feature extraction from raw signals. Ondus [5] employed a customized AlexNet to extract phase offset features, achieving high-precision classification of hovering UAVs. Ezuma et al. [7] analyzed the energy distribution of RF signals and combined k-Nearest Neighbor (KNN) with Neighborhood Component Analysis (NCA) to filter irrelevant features and improve detection accuracy in the presence of interference. Nemer et al. [8] proposed a hierarchical learning framework using four classifiers to sequentially identify the presence, type, and specific flight modes of a UAV. To enhance recognition performance for micro-UAVs in complex environments, Xu et al. [9] introduced adaptive decomposition techniques, such as Empirical Mode Decomposition (EMD) and Ensemble EMD (EEMD), to extract intrinsic mode function (IMF) features.

To overcome the scarcity of large-scale labeled data and the performance degradation caused by domain shift,

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researchers have increasingly adopted Self-Supervised Learning (SSL) and transfer learning. Li et al. [10] developed RadioNet, which utilizes adversarial learning to enhance RFF stability across cross-day deployment scenarios. The latest PRLS-RFF [11] framework introduces physics-consistent multi-view self-supervised contrastive pretraining. By employing a Mamba dual-stream backbone, the model simultaneously captures temporal and time-frequency features across transient and steady-state regimes, significantly improving cross-domain generalization and open-world device recognition.

The adaptation of pre-trained large models for wireless physical-layer tasks represents a transformative trend in the field. Zhao et al. [12] proposed a framework that integrates learnable shapelets with pre-trained LLMs. In this architecture, shapelets explicitly capture local temporal patterns, while the LLM models long-range global dependencies, thereby addressing the "black-box" interpretability challenge inherent in deep neural networks. Zheng et al. [13] developed an RFF-LLM framework based on a modified GPT-2 to extract high-dimensional temporal features. To facilitate deployment on resource-constrained devices, they utilized a Proximal Policy Optimization (PPO)-based dynamic knowledge distillation (KD) strategy to transfer the teacher model's knowledge to a lightweight Lite-HRNet, balancing identification accuracy with computational efficiency.

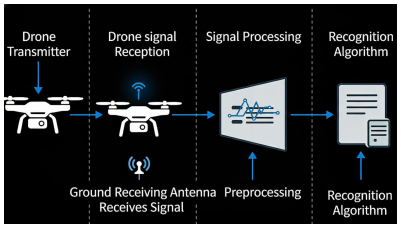
However, Large Language Models (LLMs) exhibit significant limitations in spectrum recognition tasks. Their massive parameter counts impose prohibitive storage and computational burdens, which hinders efficient deployment on resource-constrained edge devices or terminals [14]. Furthermore, the high computational complexity inherent in these models results in substantial inference latency, making it difficult to satisfy stringent real-time operational requirements in wireless environments [15]. To fulfill the rigorous imperatives of real-time operational response and high-fidelity authentication in UAVs radio frequency (RF) identification, it is essential to advance high-precision, edge-compatible recognition paradigm.

III. METHORDS

A. Overview

The core approach of this paper is converting radiated frequency signals from UAVs into image-domain information, and leveraging an advanced detector network for target recognition and detection. Key outputs include UAV type, signal frequency, and duration extracted from spectrograms, which serve as observational inputs to support subsequent operations (e.g., jamming, physical countermeasures). This study focuses on recognizing UAVs controlled via wireless links, excluding those operated through wired links such as optical fibers（As show in the Fig. 1.）

Figure 1. System Architecture UAV RF Fingerprints Recognition



B. Multi-domain Feature Mapping and Visual Representation of RF Signals

Radio Frequency (RF) signals typically manifest as one-dimensional time-domain sequences characterized by non-stationarity, high noise levels, and complex phase variations. While 1D Convolutional Neural Networks (1D-CNNs) can process raw signal data, they often struggle to capture intricate joint time-frequency representations [16]. To fully leverage the **YOLO26** [4] architecture—specifically its advanced capabilities in spatial feature extraction, multi-scale fusion, and attention mechanisms—this study proposes a transformation method based on the Short-Time Fourier Transform (STFT). This approach maps 1D RF signals into 2D images containing rich texture and intensity information, effectively reformulating the signal classification task into a computer vision object detection problem [17].

The raw RF signal consists of In-phase (I) and Quadrature (Q) components. The received discrete-time signal $x[n]$ is expressed analytically as

$$x[n] = I[n] + jQ[n], n = 0,1,\ldots,N-1 \tag{1}$$

where N denotes the number of samples. Given the complexity of wireless channels, raw signals are frequently corrupted by Gaussian white noise and multipath interference. To enhance the Signal-to-Noise Ratio (SNR) and standardize the input dynamic range, Z-score normalization is applied to the I and Q channels independently:

$$\hat{s} = \frac{s[n] - \mu_s}{\sigma_s} \tag{2}$$

where $S \in \{I, Q\}$, and μ_s , σ_s represent the mean and standard deviation of the respective components. This normalization eliminates amplitude scaling discrepancies, facilitating subsequent time-frequency transformations.

To capture the time-varying frequency characteristics of the signal, STFT is employed to map the 1D sequence into the 2D time-frequency domain. Compared to wavelet transforms, STFT maintains an optimal balance between computational efficiency and feature resolution, making it suitable for large-scale dataset construction [18, 19].

For the preprocessed signal $x[n]$, the STFT is defined as:

$$X(m,\omega) = \sum_{\infty}^{-\infty} x[n] w[n - mR] e^{-j\omega n} \tag{3}$$

where $w[n]$ is the window function (a Hamming window is selected to suppress sidelobe leakage), R is the hop size, and ω is the angular frequency.

To adapt the complex output matrix for the YOLO26 visual input layer, we extract the Power Spectral Density (PSD) and convert it to a logarithmic scale to enhance the visualization of weak signal features [20]:

$$S(m,\omega) = 10 \cdot lg(|X(m,\omega)|^2 + \epsilon) \tag{4}$$

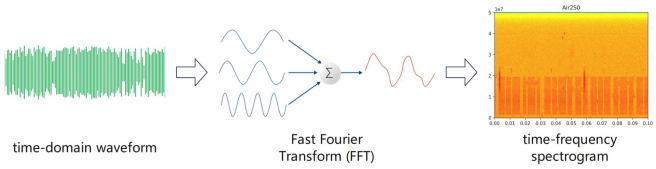
Here, ϵ is a small constant to prevent divergence during logarithmic calculation. The resulting spectrogram matrix $S \in \mathbb{R}^{H \times W}$ visually reflects the spectral evolution, where transient signal characteristics appear as distinct texture

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patterns—features that CNNs are adept at extracting.（As shown in Fig. 2）

The generated spectrogram matrix S is inherently single-channel data. To maximize the efficacy of the cross-channel feature fusion modules in the YOLO26 backbone and to utilize weights pretrained on ImageNet, the single-channel spectrogram is converted into a three-channel RGB image via pseudo-color mapping. A viridis colormap is applied to map intensity values to RGB pixel values. This mapping translates subtle intensity variations into color gradients, significantly enhancing feature contrast.

Figure 2. UAVs RF Signal Preprocessing Flowchart



B. *Network Architecture Design and Loss Optimization Mechanism of YOLO26*

While the standard YOLO framework is optimized for natural scene images, RF spectrograms exhibit unique characteristics, such as high aspect ratios, sparse texture distribution, and overlapping signal footprints. To address these challenges, this study optimizes the loss function tailored to the task-specific characteristics of RF signal detection.

YOLO26 [4] is an algorithm developed to upgrade the real-time performance based on the YOLOv11 [21] version. It retains the feature extraction capabilities of SPPF, C2SPA, C23K, and other modules in YOLOv11, while adopting NMS mode optimization, making it suitable for deployment on heterogeneous embedded platforms. UAVs RF fingerprint recognition is a typical high-real-time task that requires a balance between detection accuracy and real-time performance. Therefore, this algorithm is selected for the task in this study. (As shown in Fig. 3)

The training objective is governed by a composite loss function designed to handle the class imbalance inherent in RF datasets and the geometric distortion of spectrogram features. The total loss $\mathcal{L}_{total}$ is defined as:

$$\mathcal{L}_{total} = \lambda(t)\cdot\mathcal{L}_{cls} + (1-\lambda(t))\cdot\mathcal{L}_{box} \tag{5}$$

Classification Loss ($\mathcal{L}_{cls}$): We employ Varifocal Loss (VFL) instead of standard Cross-Entropy to prioritize hard-to-classify examples.

$$\mathcal{L}_{cls}(p,q) = \begin{cases} -q(q\log(p) + (1-q)\log(1-p)) - \alpha p^{\gamma}\log(1-p) & \text{if q>0} \\ \alpha p^{\gamma}\log(1-p) & \text{if q=0} \end{cases} \tag{6}$$

where p is the predicted score and q is the IoU-aware target score.

Box Loss($\mathcal{L}_{box}$): "To accommodate the independent temporal-spectral dimensions and extreme aspect ratios of RF spectrograms, we integrate a decoupled bounding box regression strategy into YOLO26. By replacing CIoU [22] with Efficient IoU (EIoU) [23], we independently minimize width and height discrepancies, mitigating the coupling effects of CIoU's aspect ratio penalty. This approach accelerates convergence and improves localization for RF signals with varying bandwidths and durations. The box loss is defined as:"

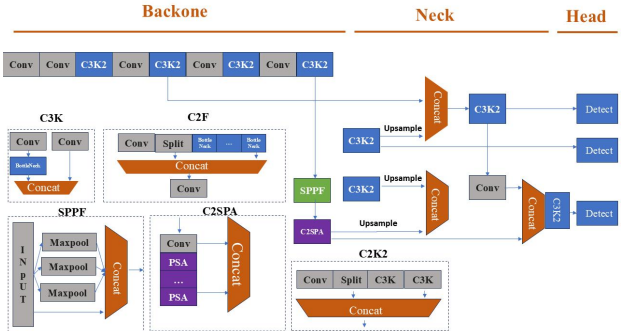
$$\mathcal{L}_{box} = 1 - \mathcal{L}_{ELoU} \tag{7}$$

In RF signal detection, the variations in signal duration (width w) and bandwidth (height h) are strictly independent physical phenomena. EIoU [23] decomposes the aspect ratio loss into two explicit difference terms:

$$\mathcal{L}_{EIoU} = 1 - IoU + \frac{\rho^2(b,b^{gt})}{c^2} + \frac{\rho^2(w,w^{gt})}{c_w^{\ 2}} + \frac{\rho^2(h,h^{gt})}{c_h^{\ 2}} \tag{8}$$

where c_w and c_h are the width and height of the smallest enclosing box covering the predicted and ground truth boxes, By directly minimizing $\frac{\rho^2(w,w^{gt})}{c_w^{\ 2}}$ and $\frac{\rho^2(h,h^{gt})}{c_h^{\ 2}}$, the EIoU loss functions as a decoupled optimizer. It allows the model to learn the time-domain features and frequency-domain features separately.

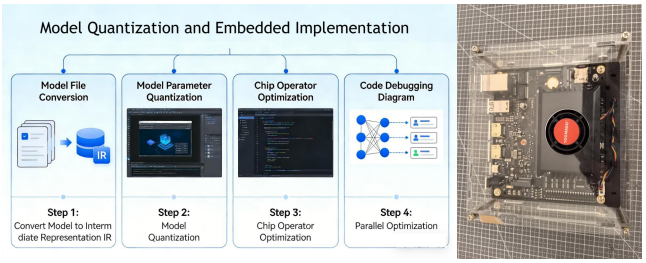
Figure 3. YOLO26 Framework



C. *Hardware Environment and Deployment*

To facilitate real-time inference on resource-constrained edge devices, precision optimization is employed to reduce the computational complexity and memory footprint of the YOLO26 architecture. In this study, we implement half-precision conversion to transform the network weights and activations from 32-bit floating-point (FP32) to 16-bit half-precision floating-point (FP16). The quantized YOLO26 model is exported to the ONNX intermediate representation (IR) and subsequently optimized for the Ascend NPU using the Huawei Compute Architecture for Neural Networks (CANN) software stack. Specifically, the model is converted into an Offline Model (.om) via the Ascend Tensor Compiler (ATC).

Figure 4. Hardware Environment and Deployment Pipeline



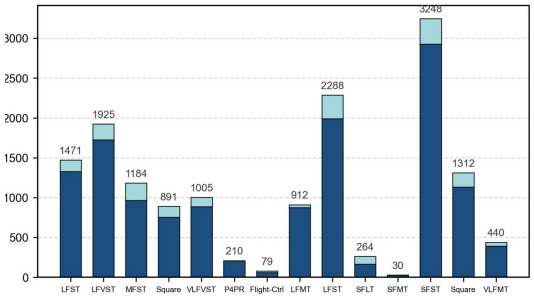
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IV. Data and Experiments

A. Dataset

RFUAV[24] builds benchmark fo RF detection. The dataset is constructed for unmanned aerial vehicle (UAV) radio frequency (RF) signal detection and identification, focusing on two core RF signal types: frequency hopping (FH) signals and image transmission (IT) signals. It contains 14 valid signal categories with a total of 15,259 samples, which are partitioned into three subsets: training, validation, and test sets, following the conventional 7:1:2 split ratio for deep learning-based signal classification tasks. Specifically, the training set includes 10,681 samples (70.0%), the validation set has 1,526 samples (10.0%), and the test set comprises 3,052 samples (20.0%).(As shown in Fig. 5)

Figure 5. Data Summary



B. Experimental Setup and Hardware Environment

All experiments were conducted on a high-performance workstation equipped with an Intel Core i7-9700K CPU and an NVIDIA GeForce RTX 4060 Ti GPU. The model implementation and training were performed using the PyTorch 2.9.1 deep learning framework. To accelerate the training process and optimize memory usage, Automatic Mixed Precision (AMP) was enabled. The model was trained using a standardized input resolution of 640 × 640 pixels to balance spatial detail with computational efficiency. The training process spanned 100 epochs with a batch size of 32, ensuring stable convergence. The proposed model is deployed on the Ascend 310B AI chip, a high-efficiency edge processor with the Da Vinci architecture. The hardware prototype of the Ascend 310B is illustrated in Fig. 4.

C. Evaluation Criteria Metrics

mAP (mean Average Precision) is used as the metric for object detection. In the field of object detection in machine learning, mAP is a crucial indicator to measure the performance of object detection algorithms(mAP50 and mAP95 as the standard evaluation metrics for detection accuracy.). Generally, the mean Average Precision (mAP) is obtained by comprehensively weighted averaging the Average Precision (AP) of all categories.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

Wbere TP (True Positive) refers to correctly identified positive samples;FP (False Positive) refers to incorrectly identified positive samples;FN (False Negative) refers to incorrectly identified negative samples.

The Average Precision (AP) is calculated based on precision and recall, which is the area under the curve with precision as the vertical axis and recall as the horizontal axis. The formula is as follows:

$$AP = \int_0^1 P(R)dR \tag{11}$$

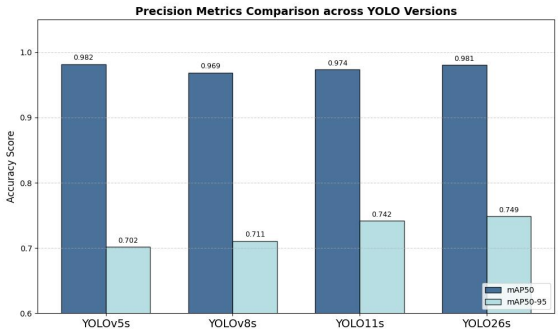
To evaluate the proposed YOLO26 [4], YOLOv5, YOLOv8, and YOLOv11 [21] are selected as representative benchmarks. YOLOv5 provides a widely recognized industrial standard, while YOLOv8 serves as a state-of-the-art (SOTA) reference with its advanced anchor-free architecture. YOLOv11, representing the latest technical frontier, is included to validate the model against the most recent advancements.

V. Results and Analysis

A. Overall Performance Comparison

This study evaluates the proposed YOLO26 against YOLOv5s, YOLOv8s, and YOLO11s, with quantitative results shown in Fig 6. For mAP50, YOLO26 scores 0.981, close to YOLOv5s (0.982) and higher than YOLOv8s (0.969) and YOLO11s (0.974). More critically, on the stricter mAP50-95 metric, YOLO26 reaches 0.749, outperforming YOLO11s (0.742) and exceeding YOLOv5s by 6.7%. While all models enable basic signal detection, YOLO26 shows stronger performance in box regression and localization. Its higher spatial sensitivity benefits accurate RF signal separation in dense spectra. Overall, YOLO26 balances accuracy and localization well, making it highly competitive for UAV RF signal detection.

Figure 6. Precision Metrics Comparison across YOLO Versions

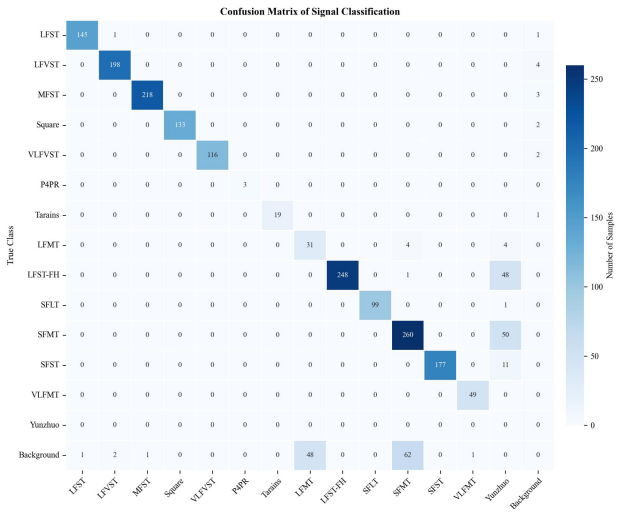


B. Class-wise Detection Analysis

The confusion matrix in Fig. 7 depicts the discriminative performance of the 14-class YOLO26 model. High diagonal concentrations for modulations such as MFST and SFMT underscore robust intra-class feature representation. While minor inter-class confusion between SFMT and 'Yunzhuo' arises from spectral overlaps, the model maintains a minimal false-negative rate across all categories. These results validate the architecture's efficacy in accurately isolating and identifying complex UAV signals within high-interference environments.

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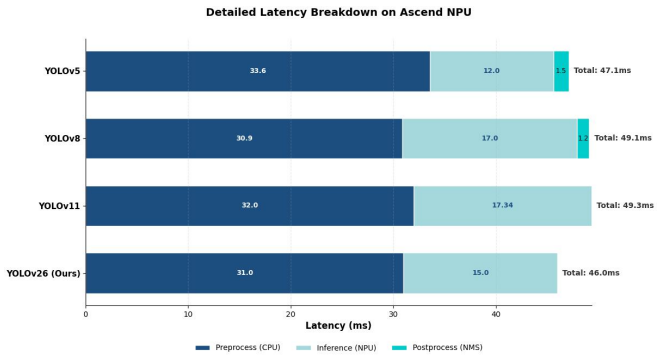
Figure 7. Confusion matrix of the proposed YOLO26 model



C. Computational Complexity and Efficiency

As illustrated in the detailed latency breakdown (Fig. 8), YOLO26 achieves the superior end-to-end inference performance with a total latency of 46.0 ms, outperforming YOLOv5 (47.1 ms), YOLOv8 (49.1 ms), and YOLOv11 (49.3 ms).Data analysis reveals that while CPU-based preprocessing constitutes a significant portion of the overhead across all models (31–33 ms), the NPU-accelerated inference phase highlights the architectural efficiency of YOLO26. Specifically, our model achieves an NPU inference latency of 15.0 ms, which is a 11.8% and 13.5% reduction compared to YOLOv8 (17.0 ms) and YOLOv11 (17.34 ms), respectively. Although YOLOv5 exhibits a lower raw NPU latency (12.0 ms), its overall execution is hindered by higher preprocessing demands. These results demonstrate that YOLO26 provides an optimal balance between algorithmic complexity and hardware-specific execution efficiency, making it highly suitable for real-time RF signal sensing on edge-computing AI platforms.

Figure 8. Performance evaluation of models on Ascend 310B



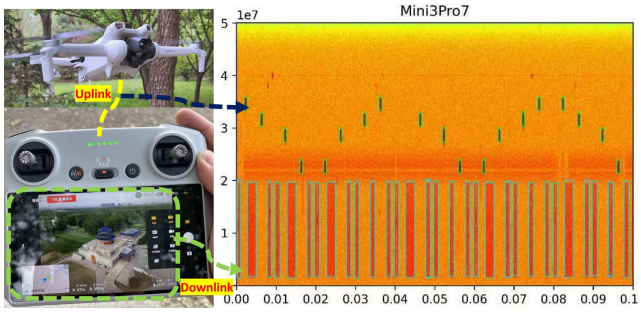
D. Qualitative Results and Visualization

1) Mini3Pro7 Case

To intuitively evaluate the detection proficiency of the proposed YOLO26, Fig. 9 presents the visualization of detection results on a representative UAVs RF signal spectrogram (Mini3Pro7). The spectrogram reveals a complex electromagnetic environment characterized by the coexistence of heterogeneous signal components. In the lower frequency band (0–20 MHz), a series of high-occupancy broadband bursts are observed, exhibiting periodic temporal patterns consistent with high-throughput video downlink transmissions. Concurrently, the upper spectral region (20–40 MHz) displays discrete, narrowband pulses following a deterministic frequency-hopping (FH) sequence, typical of command and control (C2) uplinks.

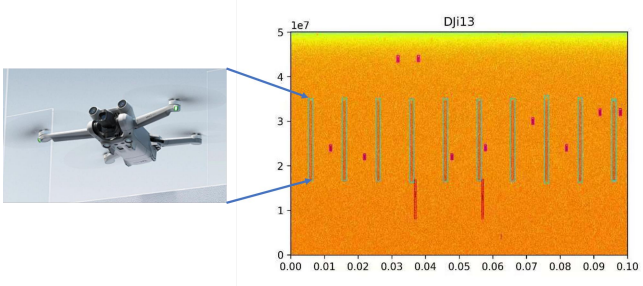
As illustrated by the predicted bounding boxes, YOLO26 demonstrates exceptional multi-scale feature extraction capabilities. It accurately encapsulates the extensive broadband segments while simultaneously isolating the transient, localized FHSS hops with high localization precision. Notably, the model maintains high confidence levels despite the varying power spectral densities and the presence of spectral leakage between adjacent pulses. This qualitative evidence substantiates that the proposed architecture effectively learns the discriminative time-frequency signatures of complex UAVs signals, ensuring robust performance in scenarios where broadband interference and transient signals overlap.

Figure 9. Mini3Pro7 Case



2) Dji13 Case

Figure 10. DJI13 Case



The generalization of YOLO26 is validated on the Dji13 data (Fig. 10), featuring coexisting wideband control signals and agile FHSS components. The model precisely localizes periodic wideband bursts (cyan) and transient, low-power FHSS hops (magenta) across a 10–45 MHz bandwidth. Notably, YOLO26 maintains accurate bounding box regressions even when FHSS signals are in close spectral proximity to dominant transmissions or partially obscured by thermal noise. This performance underscores the model's ability to extract deep semantic features from sparse

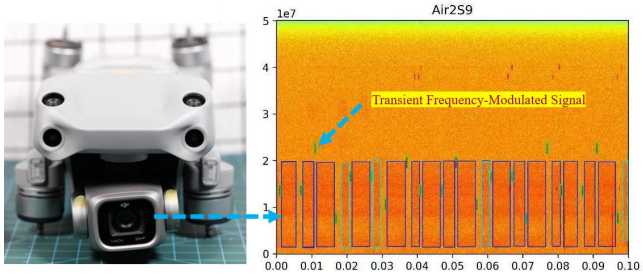
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spectrograms, confirming its robustness in monitoring multi-protocol UAV activities within high-entropy electromagnetic environments.

3) Air2S9 Case

The robustness of YOLO26 is further validated using the Air2S9 data(Fig. 11), which presents a congested spectral environment with high temporal occupancy. In the 0–20 MHz range, the model precisely segments wideband bursts separated by minimal guard intervals, a capability essential for accurate UAV downlink duty cycle estimation. Furthermore, within the 20–25 MHz band, YOLO26 effectively isolates transient, low-amplitude pulses (green boxes) despite significant sidelobe interference and spectral leakage from dominant signals. This high-fidelity localization under dense spectral conditions underscores the efficacy of the integrated multi-scale feature fusion and spatial attention mechanisms. Ultimately, these results confirm YOLO26's reliability in detecting advanced UAV protocols that utilize complex, overlapping time-frequency modulation strategies.

Figure 11. Air2S9 Case



VI. Conclusion

This study presented YOLO26, a specialized framework for RF signal detection optimized for edge-computing environments. By integrating a decoupled bounding box regression strategy using EIoU, the model effectively addresses the unique temporal-spectral characteristics and extreme aspect ratios of RF spectrograms. Experimental results demonstrate that YOLO26 achieves superior classification accuracy across 14 categories and maintains robust performance amidst significant interference. Furthermore, hardware validation on the Ascend 310B NPU confirms its computational efficiency, achieving an optimal end-to-end latency of 46.0 ms. These advancements establish YOLO26 as a highly effective and deployable solution for real-time UAV monitoring and RF signal sensing.

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